

UNIT 1 NOTES

Information Technology Application in Business

Information Technology (IT) refers to the use of computers, software, databases, networks, cloud computing, and other digital technologies to collect, process, store, retrieve, and communicate information efficiently.

In business, IT is used to support **business operations, decision-making, communication, customer service, and strategic activities**. Today, Information Technology has become an essential part of every business, helping organizations improve efficiency, enhance communication, support decision-making, and remain competitive.

Applications of IT in Business

IT supports almost every business function by improving efficiency, accuracy, and communication across departments.

- **Finance:** accounting, budgeting, financial reporting and online transactions.
- **Marketing:** digital marketing, customer analysis and advertising.
- **Sales:** order processing, sales forecasting and CRM.
- **Human Resources:** recruitment, payroll, attendance and performance management.
- **Production:** production planning, quality control and automation.
- **Inventory:** stock tracking and inventory management.
- **Customer Service:** complaint handling and customer support.
- **Management:** dashboards, reports and business analytics.

Importance of IT in Business

IT helps an organization to:

1. Automate routine activities.
2. Reduce operating costs.
3. Improve employee productivity.
4. Provide accurate and timely information.
5. Improve communication.
6. Improve customer service.
7. Support management decision-making.
8. Integrate different departments.
9. Develop new products and services.
10. Gain competitive advantage.

Real-World Example: Amazon uses Information Technology to manage online orders, digital payments, inventory, delivery tracking, and customer support through integrated information systems.

Business Enterprise Organization

A business enterprise is an organization that combines human, financial, physical, technological, and informational resources to produce goods or services and achieve organizational objectives.

An enterprise normally contains different departments, and each department performs specific functions. Although each department performs different functions, Information Technology enables them to share information efficiently and work together as an integrated system.

Main resources of an enterprise

An organization uses:

- Human resources
- Financial resources
- Physical resources
- Information
- Technology
- Materials
- Knowledge

Levels of Management

Business organizations generally have three levels of management.

1. Operational level:

Deals with routine day-to-day activities such as sales transactions, order processing and inventory management.

2. Middle management:

Monitors departmental performance and implements organizational plans.

3. Top management:

Makes strategic decisions concerning the long-term direction of the organization.

Role of IT

IT provides relevant information to every level of management, enabling operational efficiency, better monitoring, and informed strategic decision-making.

Business Functions

A business function is a specialized area within an organization responsible for performing a group of related activities that contribute to achieving business objectives

Every organization has several business functions, although their names and structures may differ.

Major Business Functions

1. Marketing

Marketing focuses on identifying customer needs and promoting products or services to satisfy those needs effectively.

Activities include:

- Market research
- Advertising
- Promotion
- Customer analysis
- Pricing

IT supports marketing through digital marketing systems, customer analytics and CRM.

2. Sales

The sales function is responsible for converting potential customers into actual buyers and generating revenue for the organization.

Activities include:

- Lead generation
- Preparing quotations
- Processing orders
- Sales forecasting
- Customer relationship management

CRM systems are commonly used to support sales activities.

3. Finance and Accounting

Finance and Accounting manage the organization's financial resources by recording transactions, controlling budgets, and preparing financial reports.

Activities include:

- Accounting

- Budgeting
- Payments
- Financial reporting
- Tax management

IT provides accounting software and ERP systems to support these activities.

4. Human Resource Management

Human Resource Management (HRM) is responsible for managing employees throughout their entire employment lifecycle, from recruitment to employee development.

Activities include:

- Recruitment
- Training
- Payroll
- Attendance
- Performance appraisal
- Employee development

HRMS and payroll systems support these functions.

5. Production/Operations

Production or Operations transforms raw materials and other resources into finished goods or services while maintaining quality and efficiency.

Activities include:

- Production planning
- Scheduling
- Manufacturing
- Quality control
- Resource management

ERP and production-management systems are commonly used.

6. Procurement

Procurement is responsible for acquiring the goods and services required for the smooth functioning of an organization.

Activities include:

- Supplier identification
- Supplier evaluation
- Purchase orders
- Receiving goods

- Invoice processing

7. Supply Chain

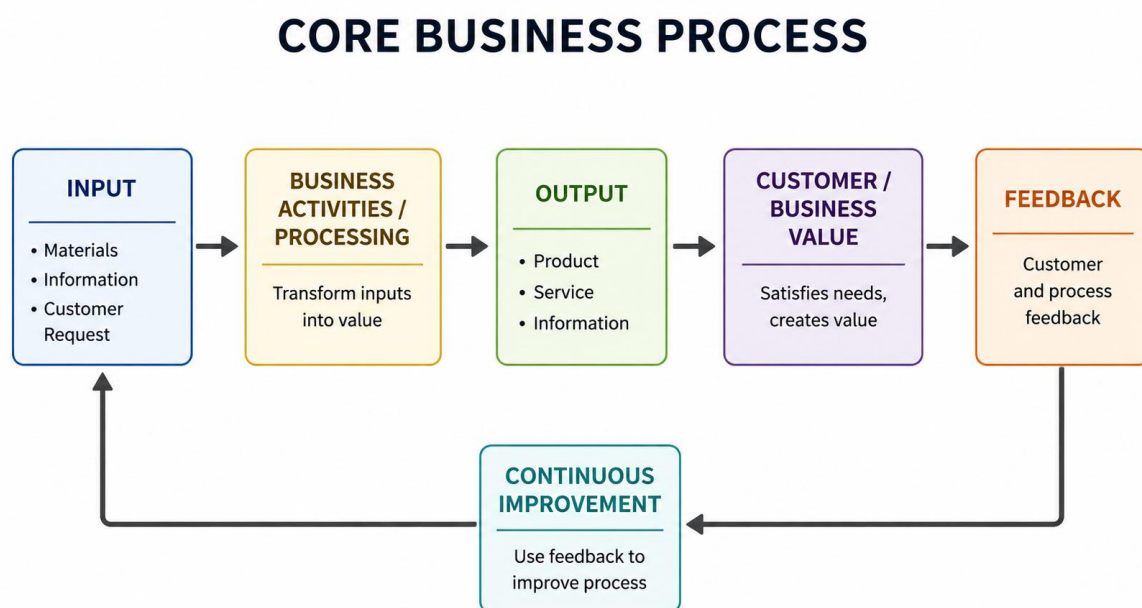
Supply Chain Management (SCM) coordinates the flow of materials, products, and information from suppliers to customers to ensure timely delivery and efficient operations.

IT supports supply-chain activities through SCM systems, inventory systems and logistics applications.

Core Business Processes

A business process is a sequence of related activities carried out to achieve a specific business objective and deliver a valuable output.

A simple business process can be represented as:



For example:

Customer order → Order processing → Product delivery → Customer

Core Business Process

Core business processes are the key activities that directly create value for customers and help an organization achieve its primary business objectives.

Important Core Processes

1. Order-to-Cash

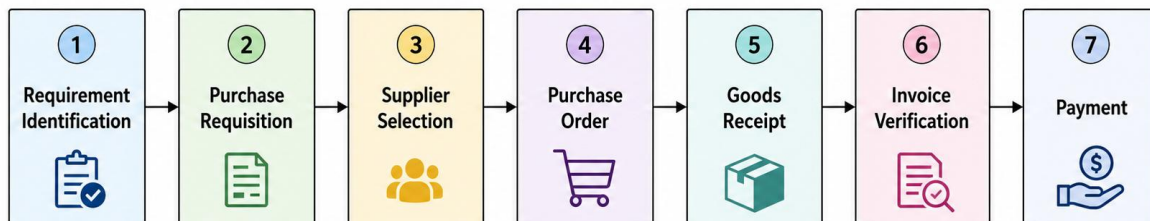
The order-to-cash process starts when a customer places an order and ends when the organization receives payment.



It involves sales, inventory, logistics and finance.

2. Procure-to-Pay

This process starts when an organization identifies a requirement for goods or services and ends when the supplier is paid.



It involves procurement, inventory and finance.

3. Hire-to-Retire

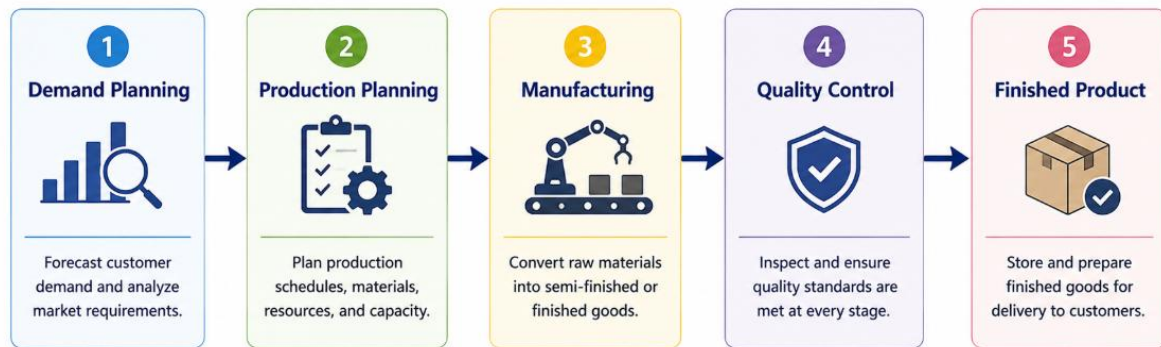
This process manages the complete employee lifecycle, from recruitment and hiring to retirement or exit from the organization.



HR systems support this entire process.

4. Plan-to-Produce

This process converts production requirements into finished goods.



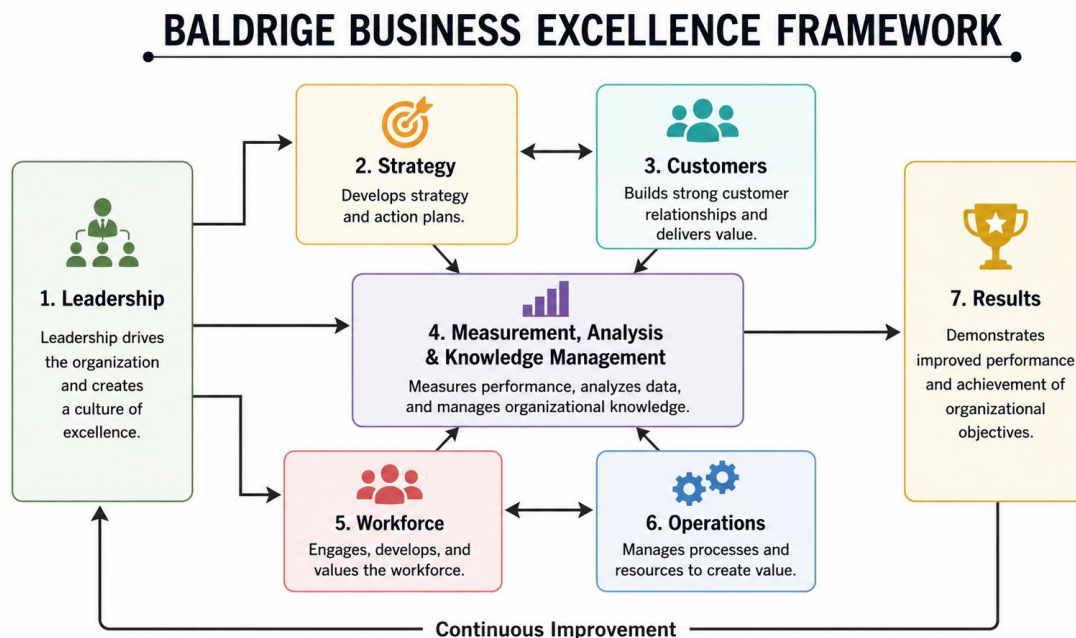
Importance of Business Processes

Well-designed processes help organizations:

- Reduce cost.
- Reduce errors.
- Improve quality.
- Reduce processing time.
- Improve customer satisfaction.
- Increase productivity.

IT helps automate, integrate, and optimize business processes, making them faster and more efficient.

Baldrige Business Excellence Framework



The Baldrige Business Excellence Framework is a management framework that helps organizations assess, manage, and continuously improve their overall performance.

It provides a systematic approach to organizational excellence. Instead of looking at departments separately, it encourages the organization to view **leadership, strategy, customers, workforce, processes, knowledge and results as interconnected areas.**

Major Categories

The framework includes:

1. Leadership
2. Strategy
3. Customers
4. Measurement, Analysis and Knowledge Management
5. Workforce
6. Operations/Process Management
7. Results

The first six areas describe how an organization operates, while **Results** examines what the organization achieves.

Leadership

Leadership refers to the ability of senior leaders to guide the organization, establish its vision and values, communicate direction, and create an environment that supports organizational success.

Leaders are responsible for ensuring that employees understand organizational objectives and work toward common goals.

Important areas of Leadership

- Establishing vision and values.
- Setting organizational direction.
- Promoting ethical behavior.
- Communicating with employees.
- Focusing on customers.
- Encouraging innovation.
- Monitoring organizational performance.
- Managing risks.
- Supporting continuous improvement.

Role of IT in Leadership

IT provides leaders with accurate and timely information for decision-making.

For example, an executive dashboard may display:

- Sales
- Revenue
- Profit
- Customer satisfaction
- Employee performance
- Inventory

Thus, IT enables leaders to make informed and fact-based decisions.

Strategic Planning

Strategic planning is the process of defining an organization's long-term objectives and determining the strategies required to achieve them..

It provides direction to the organization and helps managers decide how resources should be used.

Strategic Planning Process



Major activities

- Analyze the current business environment.
- Identify strengths and weaknesses.
- Identify opportunities and threats.
- Set strategic objectives.
- Develop strategies.
- Allocate resources.
- Implement action plans.
- Monitor results.

Role of IT in Strategic Planning

IT plays a vital role in supporting modern business strategies.

Examples include:

- E-commerce.

- Digital transformation.
- Cloud computing.
- Data analytics.
- Artificial intelligence.
- Business automation.

Real-World Example: Netflix adopted online streaming as part of its business strategy, enabling it to reach millions of customers worldwide and replace its traditional DVD rental service.

Customer Focus

Customer focus is the process of understanding customer needs, expectations, and preferences to deliver better products and services.

An organization cannot achieve long-term success without understanding and satisfying its customers.

Important activities

- Identifying customers.
- Understanding customer requirements.
- Collecting customer feedback.
- Measuring satisfaction.
- Managing complaints.
- Studying customer behavior.
- Building customer relationships.

Role of IT

IT helps organizations maintain and analyze customer information.

A **CRM system** can store:

- Customer details.
- Purchase history.
- Complaints.
- Preferences.
- Service requests.
- Communication history.

Benefits

Customer-focused IT systems help organizations:

- Respond faster.

- Personalize services.
- Improve customer satisfaction.
- Increase customer retention.
- Identify sales opportunities.

Measurement, Analysis and Knowledge Management

This category focuses on collecting, analyzing, and managing organizational information to support effective decision-making and continuous improvement.

Organizations cannot improve what they do not measure.

A. Measurement

Measurement involves collecting data about organizational performance.

Examples of **Key Performance Indicators (KPIs)** include:

- Sales revenue
- Profit
- Customer satisfaction
- Delivery time
- Product defects
- Employee turnover
- Inventory level

B. Analysis

Analysis means examining collected data to identify trends, problems and opportunities.

For example, if customer complaints have increased, analysis can identify the reason for the increase.

IT tools such as:

- Business Intelligence
- Dashboards
- Data analytics
- Data warehouses
- Reporting systems

help managers analyze information.

C. Knowledge Management

Knowledge Management (KM) is the process of creating, capturing, storing, sharing and applying organizational knowledge.

Knowledge may include:

- Procedures
- Manuals
- Employee experience
- Lessons learned
- Reports
- Technical information

IT supports knowledge management through document-management systems, knowledge bases, intranets and collaboration systems.

Workforce Focus

Workforce focus refers to how an organization manages, develops, motivates, and supports its employees to achieve organizational goals.

Employees are an important organizational resource because their skills and knowledge directly affect business performance.

Major activities

- Workforce planning.
- Recruitment.
- Employee selection.
- Training.
- Skill development.
- Performance management.
- Employee engagement.
- Recognition and rewards.
- Workplace safety.

Role of IT

IT supports workforce management through:

HRMS: Employee records and HR activities.

Payroll systems: Salary and payment processing.

Learning Management Systems: Employee training.

Recruitment systems: Job applications and hiring.

Performance systems: Employee objectives and performance evaluation.

Benefits

IT reduces administrative work, improves employee information management, supports training, and provides workforce analytics for better decision-making.

Process Management

Process management is the systematic approach to designing, monitoring, controlling, and continuously improving business processes to achieve better organizational performance.

Its primary objective is to make business processes more efficient, effective, and customer-focused

Process Management Activities

1. Identify the process.
2. Define process objectives.
3. Identify inputs and outputs.
4. Assign responsibilities.
5. Measure performance.
6. Identify problems.
7. Improve the process.
8. Monitor the improved process.

IT Support

IT can improve processes through:

- ERP.
- Workflow systems.
- Business Process Management (BPM).
- Robotic Process Automation (RPA).
- Electronic approval systems.
- Digital forms.

Example

Traditional purchase approval:

Employee → Paper Form → Manager → Finance → Procurement

Automated process:

Online Request → Electronic Approval → Procurement → Finance

The automated process reduces paperwork and processing time.

Key Purpose of Using IT in Business

The primary purpose of using Information Technology (IT) in business is to help organizations achieve their objectives more efficiently, effectively, and competitively.

Beyond storing information, IT transforms the way organizations operate, communicate, and make decisions..

Major purposes

1. Automation:

IT automates repetitive and routine activities.

2. Efficiency:

Activities can be completed faster and with fewer resources.

3. Cost Reduction:

Automation and better resource management can reduce operational costs.

4. Better Decision-Making:

Managers receive timely and accurate information.

5. Integration:

IT connects different departments and processes.

6. Better Customer Service:

Customers can receive faster and more personalized services.

7. Innovation:

IT enables e-commerce, mobile applications, digital payments and other new services.

8. Competitive Advantage:

Organizations can use technology to provide better quality, lower cost or faster service than competitors.

9. Communication:

Email, messaging, video conferencing and collaboration systems improve communication.

10. Business Control:

Managers can monitor business performance using reports and dashboards.

Enterprise Applications

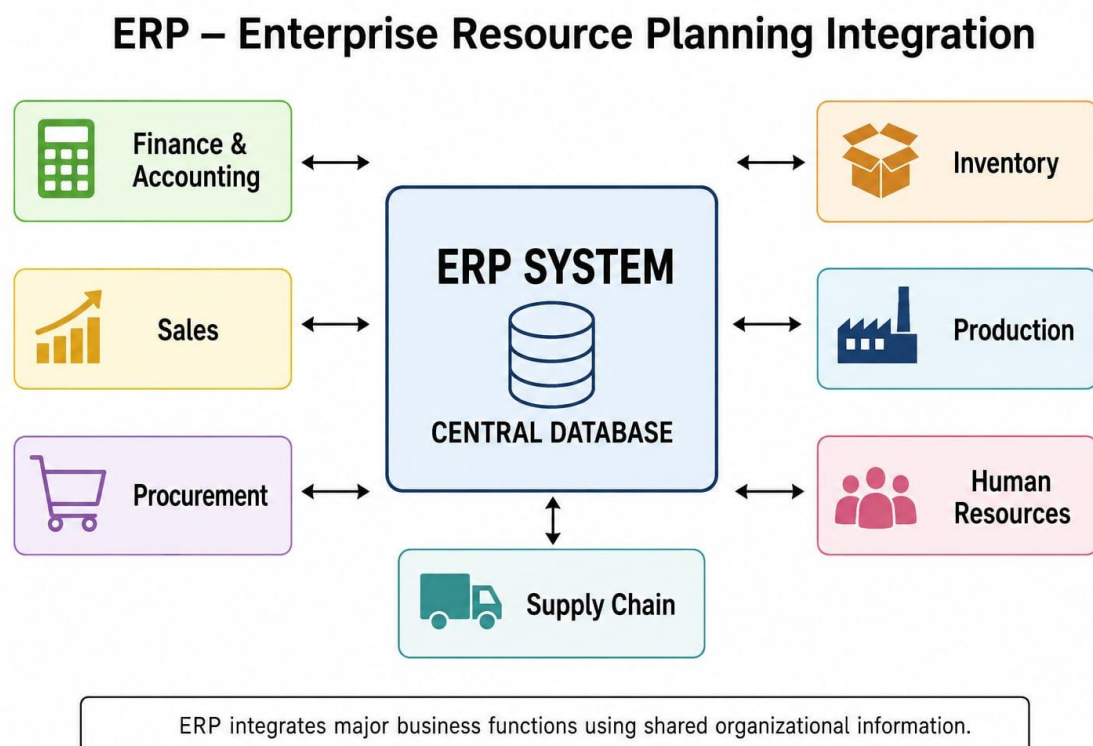
Enterprise applications are large-scale software systems designed to support critical business operations and integrate multiple departments within an organization.

Unlike small standalone applications, enterprise applications generally support multiple departments and large amounts of organizational data.

Major Enterprise Applications

- ERP — Enterprise Resource Planning
- CRM — Customer Relationship Management
- SCM — Supply Chain Management
- HRMS — Human Resource Management System
- BI — Business Intelligence

ERP — Enterprise Resource Planning



Enterprise Resource Planning (ERP) is an integrated software system that connects major business functions through a centralized database, enabling different departments to share information efficiently.

ERP can integrate:

Finance + Sales + Inventory + Procurement + Production + HR

Common ERP Modules

- Finance and Accounting
- Sales
- Procurement
- Inventory
- Production
- Human Resources
- Supply Chain

Example

Suppose a customer purchases a product.

The ERP system can:

1. Record the customer order.
2. Check inventory.
3. Update stock.
4. Generate delivery information.
5. Generate an invoice.
6. Record the financial transaction.

Thus, a single business transaction can update information across several departments.

Advantages of ERP

- Integration of departments.
- Centralized data.
- Reduced data duplication.
- Better reporting.
- Improved coordination.
- Standardized business processes.
- Better management control.

Limitations

- High implementation cost.
- Complex implementation.
- Employee training required.
- Organizational changes may be required.

CRM — Customer Relationship Management

Customer Relationship Management (CRM) is an information system used to manage and strengthen relationships with current and potential customers.

CRM is mainly focused on the **customer side of the business**.

CRM Activities

Marketing:

- Campaign management.
- Customer segmentation.

Sales:

- Lead management.
- Sales opportunities.
- Sales forecasting.

Customer Service:

- Complaints.
- Support requests.
- Service history.

Benefits of CRM

- Better customer information.
- Improved customer service.
- Faster complaint handling.
- Better sales management.
- Personalized marketing.
- Increased customer retention.

Real-World Example: Amazon uses CRM and customer data to recommend products based on previous purchases, browsing history, and customer preferences.

ERP vs CRM

Basis	ERP	CRM
Full Form	Enterprise Resource Planning	Customer Relationship Management
Purpose	Manage business operations	Manage customer relationships
Focus	Internal processes	Customers and sales
Departments	Finance, HR, Inventory, Production	Sales, Marketing, Customer Service
Users	Employees & Managers	Sales & Support Teams
Features	Planning, Inventory, Accounting	Leads, Contacts, Customer Support
Benefits	Improves efficiency	Improves customer satisfaction
Examples	SAP ERP, Oracle ERP	Salesforce, Zoho CRM

SCM — Supply Chain Management

Supply Chain Management (SCM) is the process of managing the flow of materials, products, information, and resources from suppliers to customers efficiently.

Supply Chain

Supplier → Procurement → Production → Warehouse → Distribution → Customer

IT support

SCM systems help with:

- Supplier management.
- Purchasing.
- Inventory.
- Demand forecasting.
- Warehousing.
- Transportation.
- Delivery tracking.

Benefits

- Better inventory management.
- Lower inventory costs.
- Better supplier coordination.
- Faster delivery.
- Improved demand planning.

HRMS — Human Resource Management System

A Human Resource Management System (HRMS) is an information system that manages employee records and supports various human resource activities throughout the employee lifecycle.

Functions

- Employee records.
- Recruitment.
- Attendance.
- Leave.
- Payroll.
- Training.
- Performance management.
- Employee benefits.

Benefits

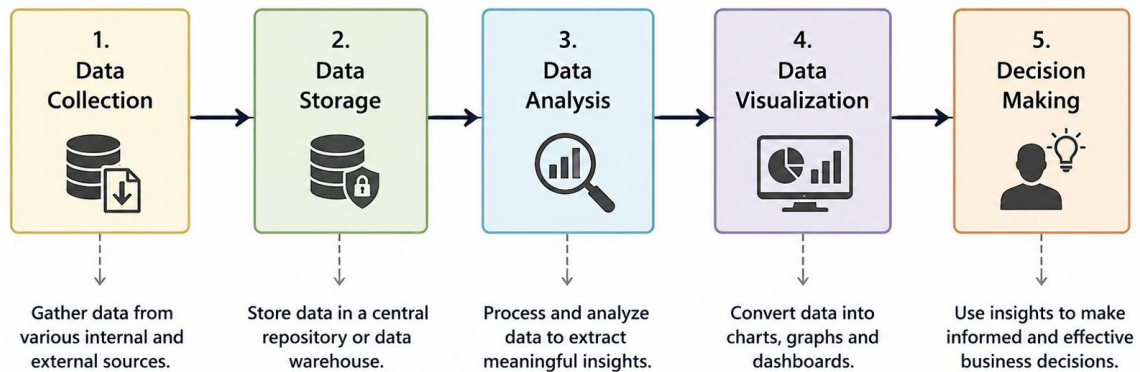
HRMS reduces manual paperwork, improves employee information management and provides HR managers with useful reports.

Real-World Example: Flipkart uses Supply Chain Management (SCM) systems to manage warehouses, coordinate deliveries, and track customer orders efficiently.

Business Intelligence

Business Intelligence (BI) refers to the technologies, tools, and processes used to collect, analyze, and present business data for informed decision- making.

Business Intelligence Process



Example

A sales manager can use a BI dashboard to determine:

- Which product sells the most?
- Which region has the highest sales?
- Which month has the lowest sales?
- Which sales employee has the highest performance?

BI therefore converts large amounts of data into useful business information.

Real-World Example: Swiggy uses Business Intelligence (BI) dashboards to analyze customer orders, identify popular food items, monitor delivery performance, and support business decisions.

Bespoke IT Application

A Bespoke IT Application is software developed specifically to meet the unique requirements of a particular organization.

It is also called:

- Custom software
- Tailor-made software
- Customized application

Unlike standard enterprise software, bespoke software is designed around the organization's specific requirements.

Example

A company may have a unique production process that cannot be properly handled by a standard ERP system. It may therefore develop a custom production-management application.

Advantages of Bespoke IT Applications

1. Exact Requirements

The software can be designed according to the organization's specific needs.

2. Flexibility

The organization can request specific features and workflows.

3. Competitive Advantage

Unique software functionality can help an organization differentiate itself.

4. Greater Control

The organization has more control over features and development.

5. Integration

The application can be designed to integrate with existing systems.

Disadvantages of Bespoke IT Applications

1. High Cost

Developing software from scratch can be expensive.

2. Longer Development Time

Requirement analysis, design, coding and testing can take considerable time.

3. Maintenance

The organization may have to manage maintenance and upgrades.

4. Development Risk

The final application may not completely satisfy business requirements.

5. Skilled Staff Required

The organization may need developers and technical specialists.

Enterprise Application vs Bespoke Application

Basis	Enterprise Application	Bespoke Application
Definition	Standard software for business processes	Software developed for a specific organization
Development	Already developed	Developed according to requirements
Users	Many organizations	Usually one organization
Customization	Limited/controlled	Very high
Implementation	Generally faster	Generally slower
Cost	License and implementation cost	Development and maintenance cost
Risk	Generally lower	Generally higher
Support	Usually vendor support	Organization/developer support
Best suited for	Standard processes	Unique processes
Examples	ERP, CRM, SCM	Custom business application

Relationship Between IT and Business Excellence

Information Technology (IT) plays a vital role in achieving business excellence by improving decision-making, streamlining business processes, strengthening customer relationships, and supporting continuous organizational improvement.

Leadership

→ IT provides dashboards, reports, and real-time information to support effective leadership and decision-making.

Strategic Planning

→ IT enables organizations to develop and implement business strategies through digital technologies, data analytics, and business intelligence.

Customer Focus

→ CRM systems help organizations understand customer needs, improve customer service, and build long-term relationships.

Measurement and Analysis

→ Business Intelligence (BI), analytic, and reporting tools help measure organizational performance and support data-driven decisions.

Knowledge Management

→ IT captures, stores, organizes, and shares organizational knowledge, making information easily accessible across the enterprise.

Workforce Focus

→ HRMS and Learning Management Systems (LMS) support employee management, training, performance evaluation, and skill development.

Process Management

→ ERP systems, workflow automation, and Business Process Management (BPM) improve business processes by increasing efficiency and reducing errors.

Results

→ IT helps organizations monitor performance, evaluate outcomes, and achieve continuous improvement through accurate and timely information.